Intel 80186, 80286 Microprocessor

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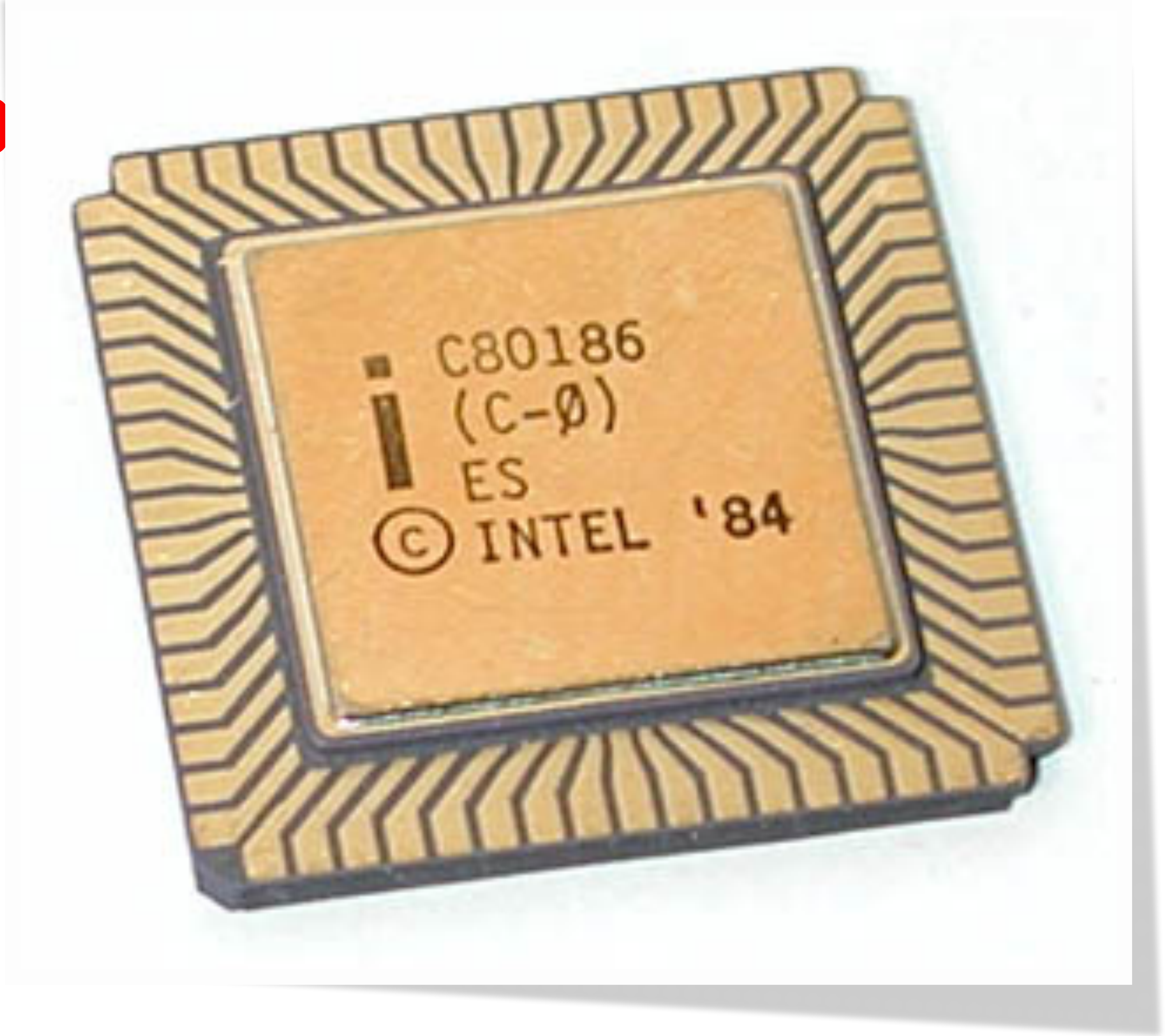
INTEL 80



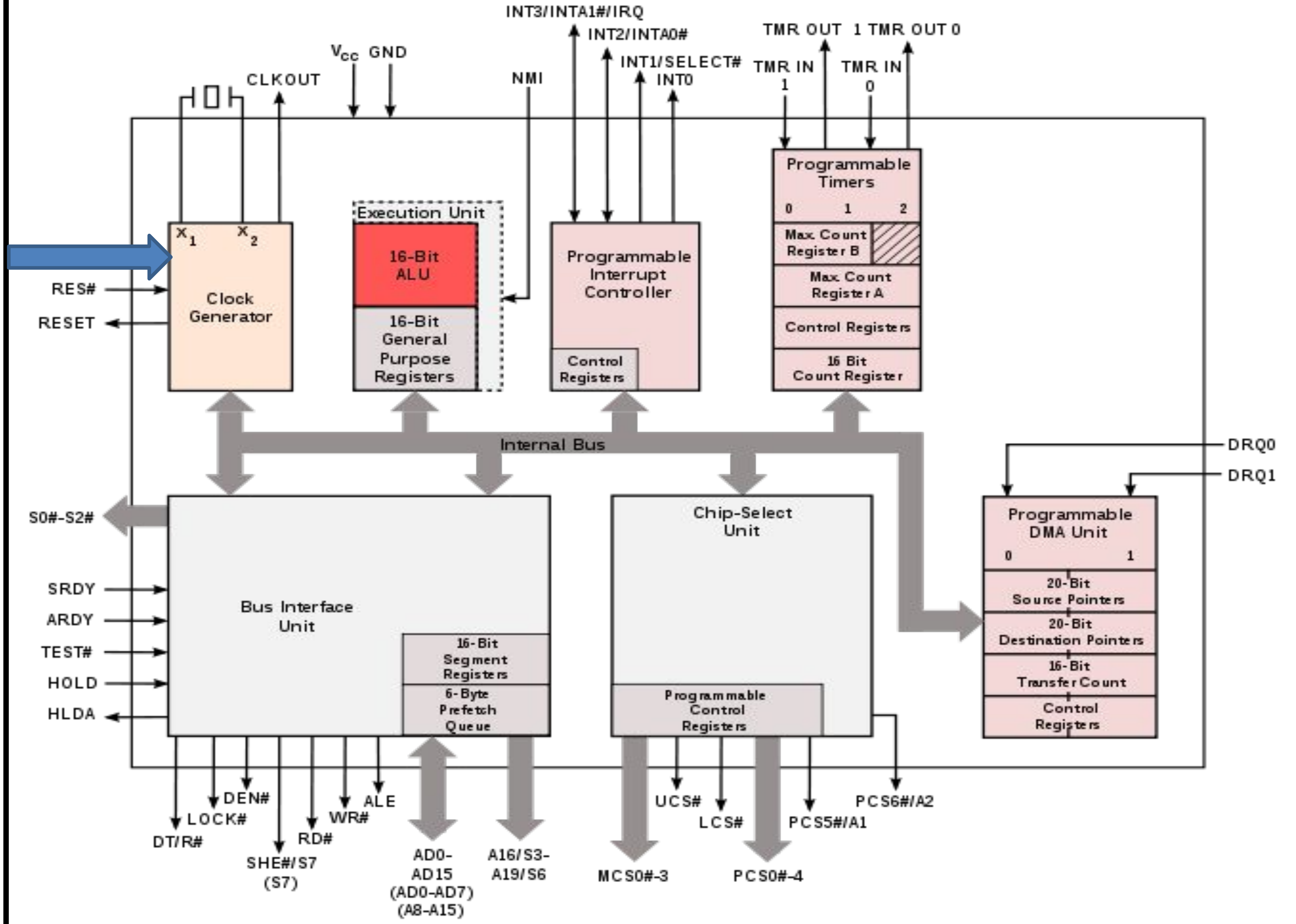
INTEL 80186

BASIC BLOCK DIAGRAM

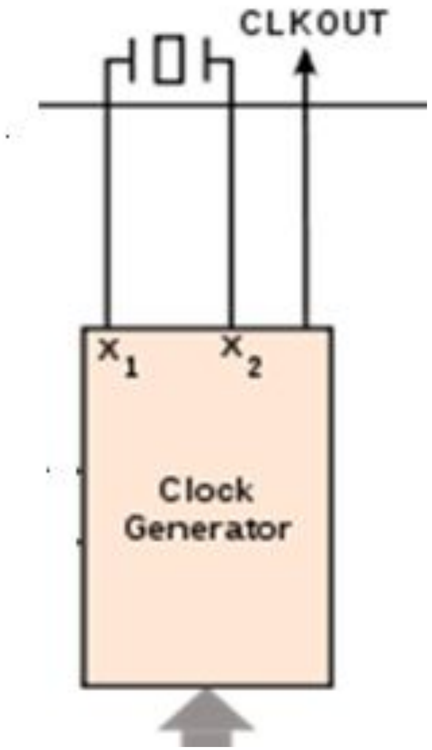
IN ADDITION TO THE BIU AND EU 80186/80188 FAMILY CONTAINS

- A CLOCK GENERATOR,
- A PROGRAMMABLE INTERRUPT CONTROLLER
- PROGRAMMABLE TIMERS
- A PROGRAMMABLE DMA CONTROLLER
- A PROGRAMMABLE CHIP SELECTION UNIT.

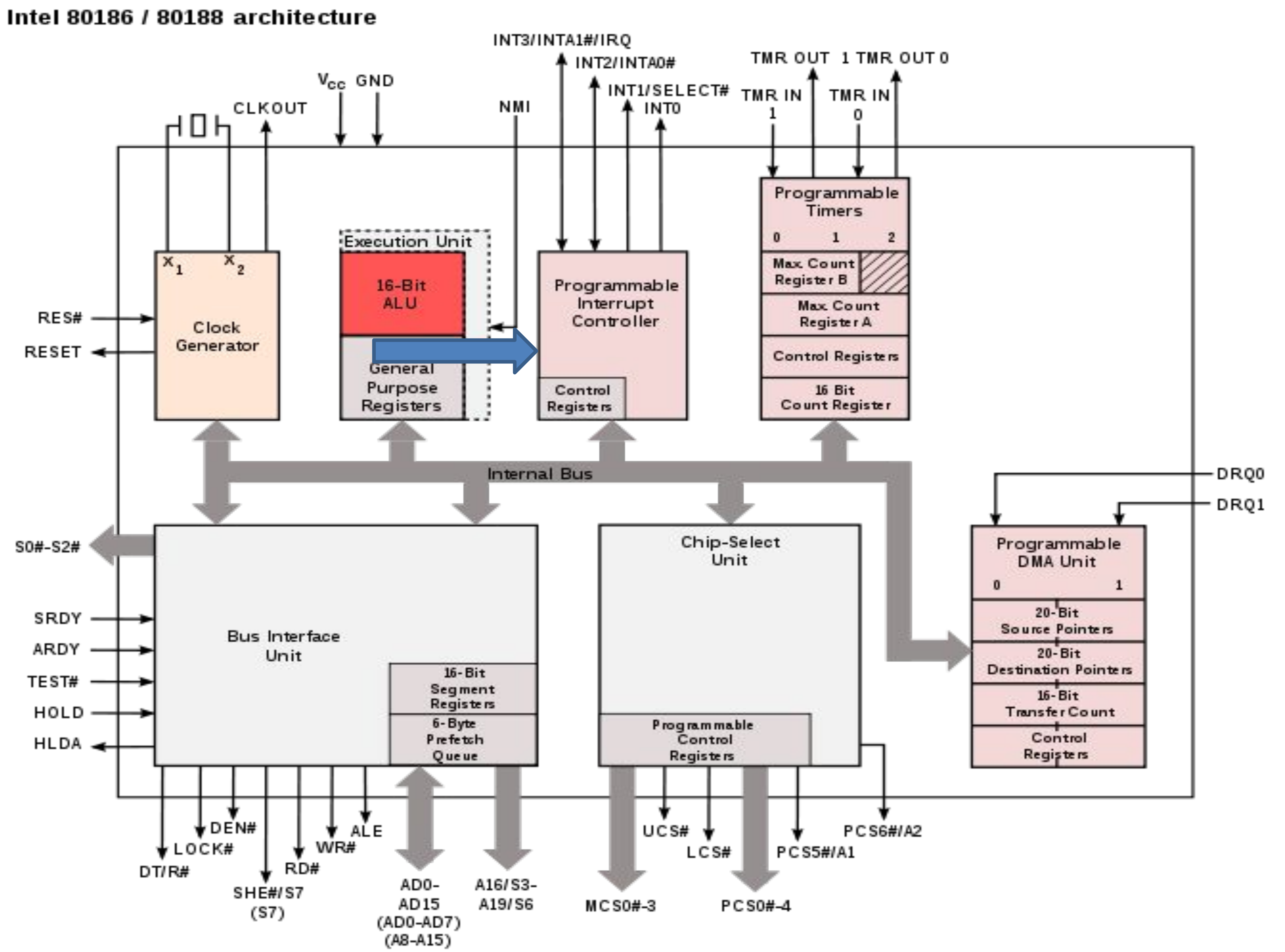
Intel 80186 / 80188 architecture



CLOCK GENERATOR

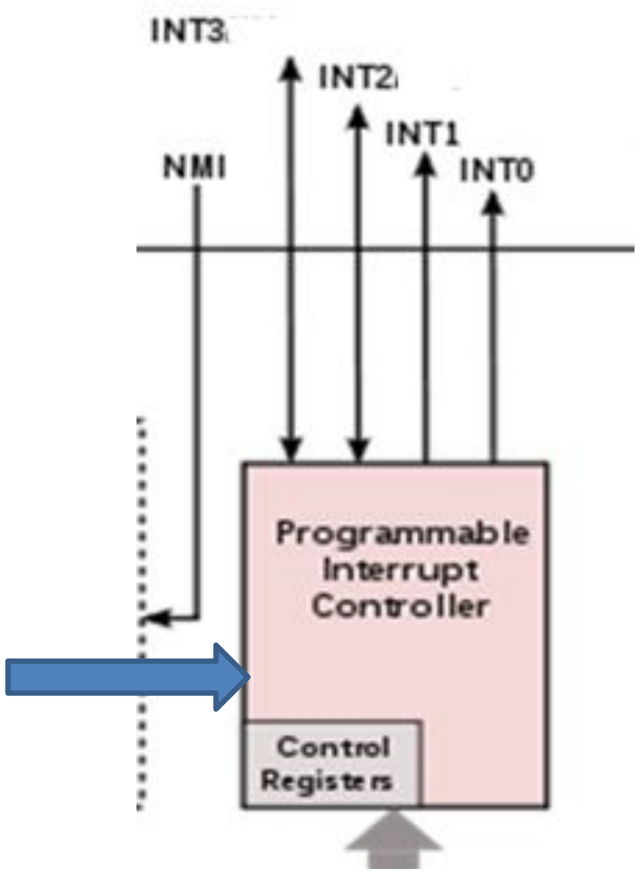


- X1,X2 CONNECTED TO CRYSTAL
- CLKOUT PROVIDS SYSTEM CLOCK SIGNAL



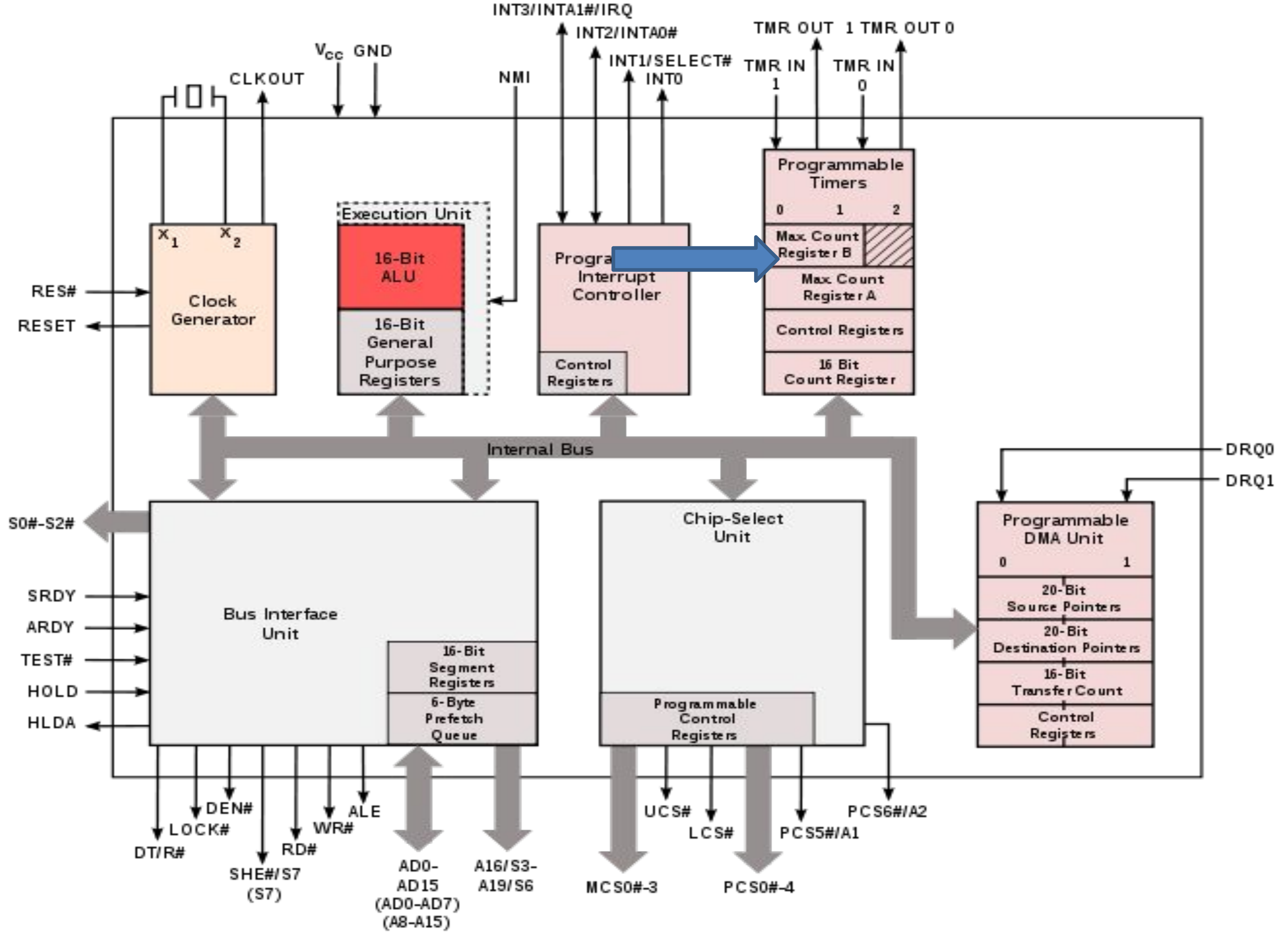
Intel 80186 / 80188 architecture
INT3/INTA1#/IRQ
INT2/INTA0#
INT1/SELECT#
INT0
NMI
TMR OUT 1
TMR OUT 0
TMR IN 1
TMR IN 0
V_{CC}
GND
CLKOUT
X_1
X_2
Clock Generator
Execution Unit
16-Bit ALU
General Purpose Registers
Programmable Interrupt Controller
Control Registers
Programmable Timers
0 1 2
Max Count Register B
Max Count Register A
Control Registers
16 Bit Count Register
RES#
RESET
Internal Bus
DRQ0
DRQ1
S0#-S2#
Chip-Select Unit
Programmable DMA Unit
0 1
20-Bit Source Pointers
20-Bit Destination Pointers
16-Bit Transfer Count
Control Registers
SRDY
ARDY
TEST#
HOLD
HLDA
Bus Interface Unit
16-Bit Segment Registers
6-Byte Prefetch Queue
Programmable Control Registers
DEN#
LOCK#
DT/R#
SHE#/S7
(S7)
RD#
WR#
ALE
AD0-AD15
(AD0-AD7)
(A8-A15)
A16/S3-A19/S6
MCS0#-3
UCS#
LCS#
PCS0#-4
PCS5#/A1
PCS6#/A2

PROGRAMMABLE INTERRUPT CONTROLLER

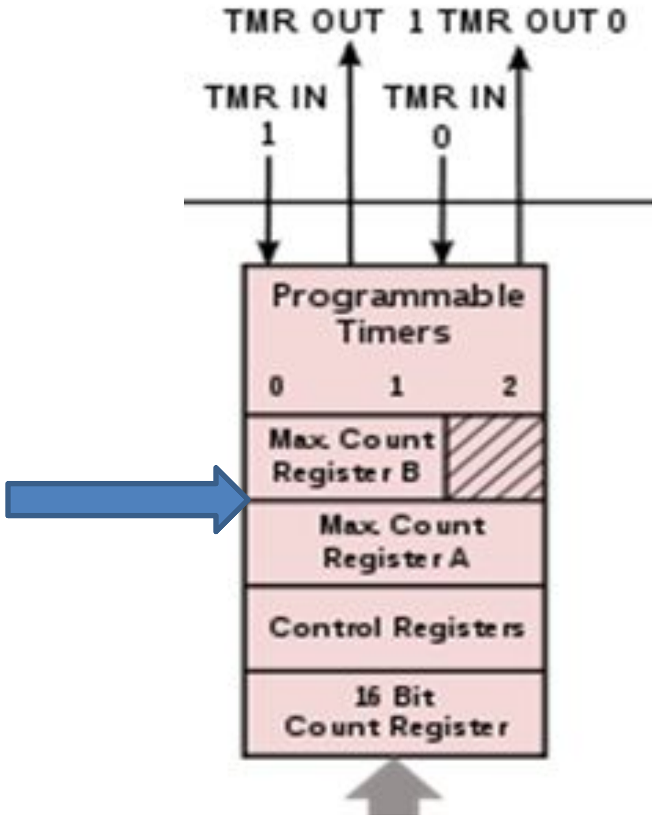


- WHICH ARBITRATES INTERNAL AND EXTERNAL INTERRUPTS

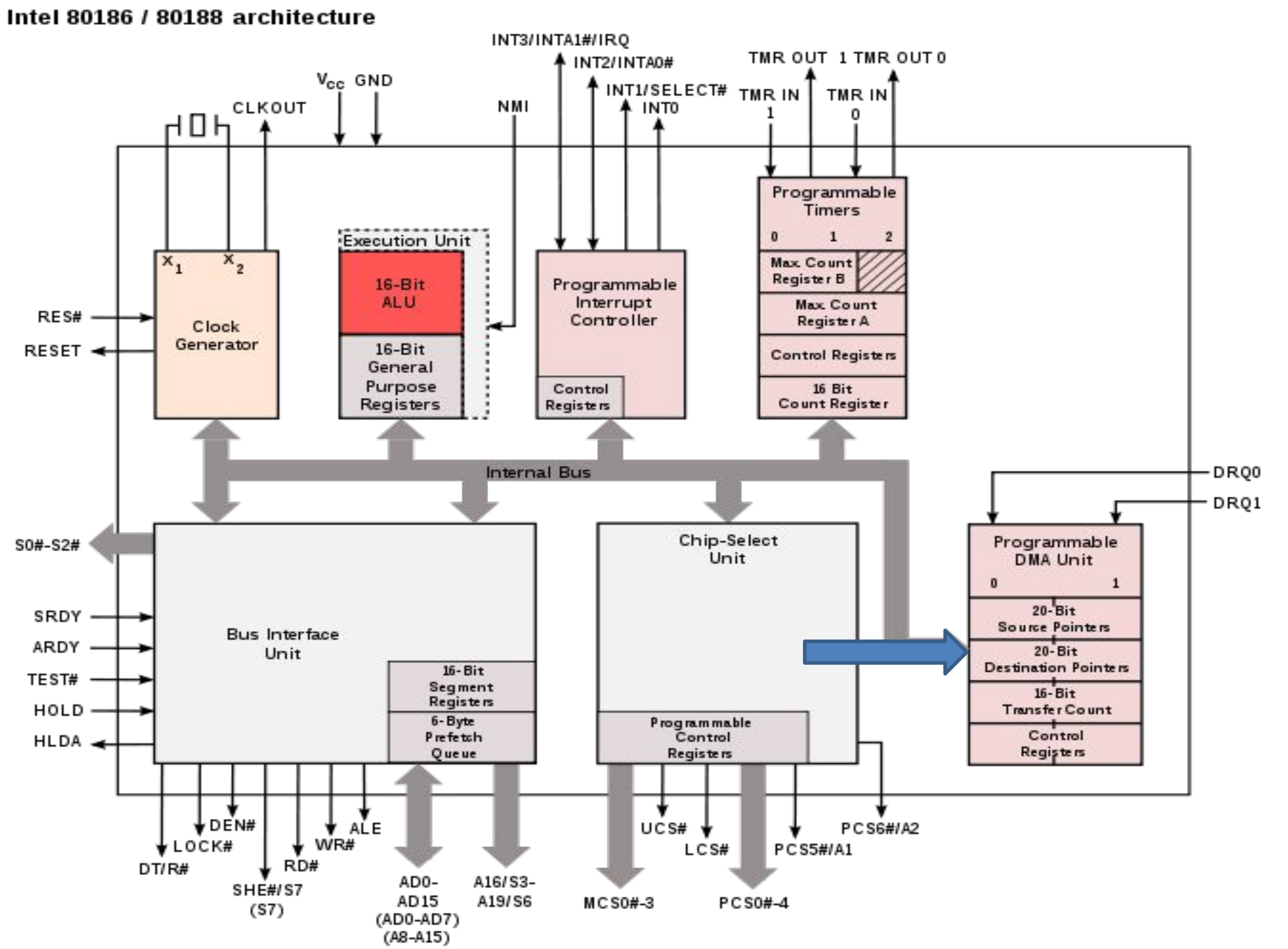
Intel 80186 / 80188 architecture



PROGRAMMABLE TIMERS



- **TIMER 0 AND 1 FOR EXTERNAL USE**
- **TIMER 2 WATCH DOG TIMER-** *It can provide a clock to the other timers*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TMR OUT 1
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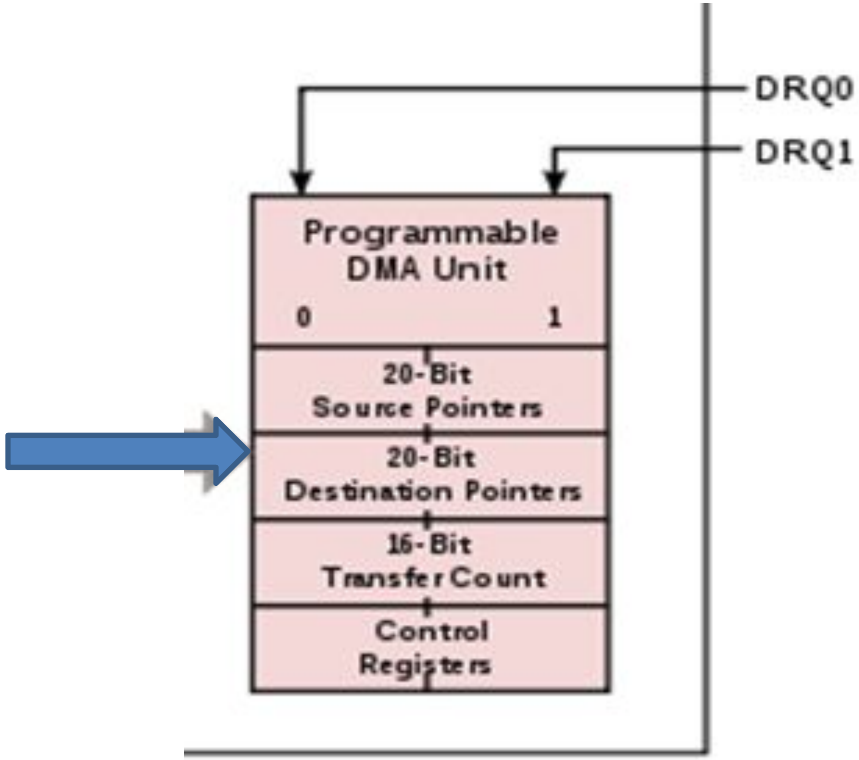
V_{cc} GND
CLKOUT
NMI
Execution Unit
16-Bit ALU
16-Bit General Purpose Registers
X_1
X_2
Clock Generator
RES#
RESET
Programmable Interrupt Controller
Control Registers
Programmable Timers
0 1 2
Max Count Register B
Max Count Register A
Control Registers
16 Bit Count Register
Internal Bus
DRQ0
DRQ1
S0#-S2#
Bus Interface Unit
16-Bit Segment Registers
6-Byte Prefetch Queue
Chip-Select Unit
Programmable Control Registers
Programmable DMA Unit
0 1
20-Bit Source Pointers
20-Bit Destination Pointers
16-Bit Transfer Count
Control Registers
SRDY
ARDY
TEST#
HOLD
HLDA
DT/R#
LOCK#
DEN#
SHE#/S7 (S7)
RD#
WR#
ALE
AD0-AD15 (AD0-AD7) (A8-A15)
A16/S3-A19/S6
MCS0#-3
UCS#
LCS#
PCS0#-4
PCS5#/A1
PCS6#/A2

PROGRAMMABLE DMA UNIT



TO TRANSFER DATA WITHOUT GOING THROUGH CPU

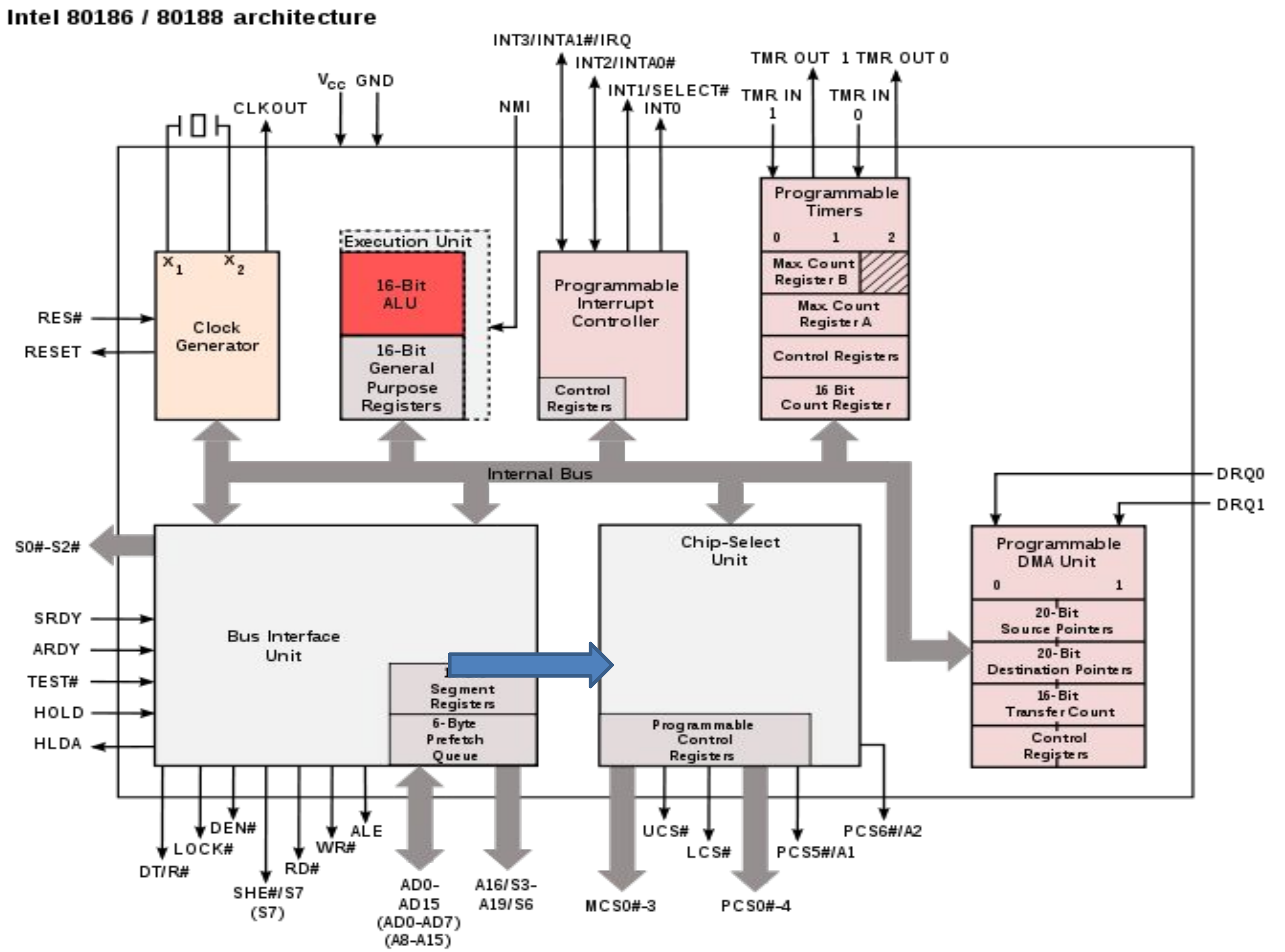
LIKE 8237

IT CAN TRANSFER DATA

-BETWEEN MEMORY LOCATIONS,

-BETWEEN MEMORY AND I/O,

OR BETWEEN I/O DEVICES.



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Segment Registers
6-Byte Prefetch Queue
Chip-Select Unit
Programmable Control Registers
Programmable DMA Unit
0 1
20-Bit Source Pointers
20-Bit Destination Pointers
16-Bit Transfer Count
Control Registers
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ARDY
TEST#
HOLD
HLDA
DT/R#
LOCK#
DEN#
SHE#/S7 (S7)
RD#
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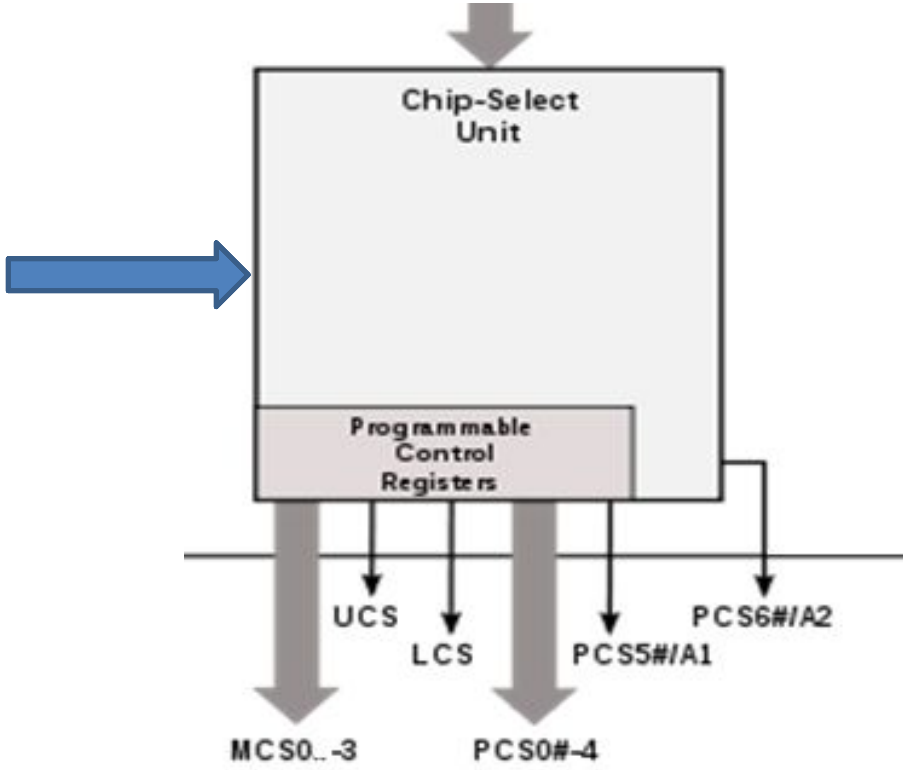
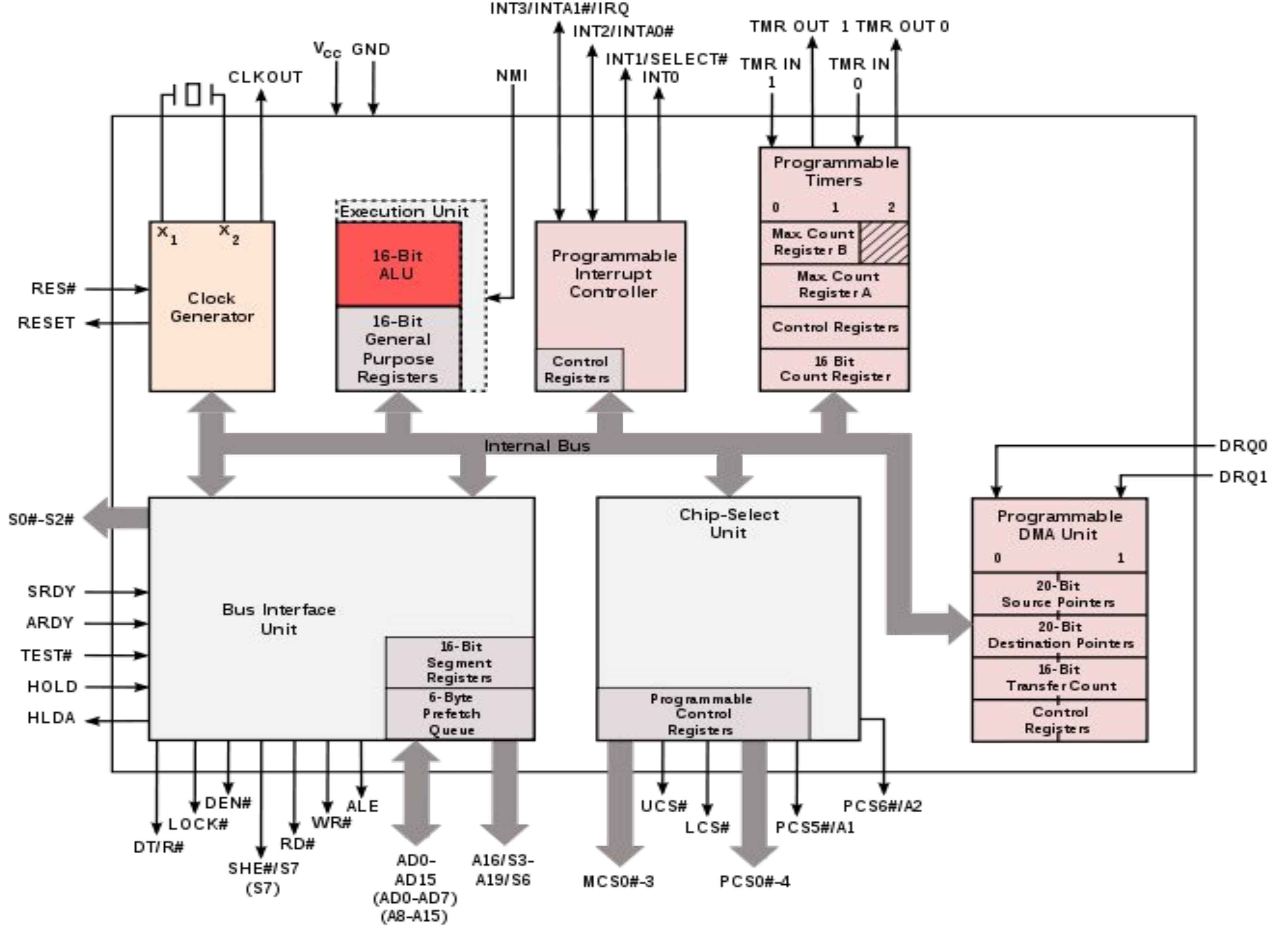
ALE
AD0-AD15 (AD0-AD7) (A8-A15)
A16/S3-A19/S6
MCS0#-3
UCS#
LCS#
PCS0#-4
PCS5#/A1
PCS6#/A2

PROGRAMMABLE CHIP SELECT UNIT



- PROGRAMMABLE BUILT IN MEMORY AND I/O DECODER

Intel 80186 / 80188 architecture



The 10 additional instructions that the 80186 has are as follows:

ENTER — Enter a procedure

LEAVE — Leave a procedure

BOUND — Check if an array index in a register is in range of array

INS — Input string byte or string word

OUTS — Output string byte or string word

PUSHA — Push all registers on stack

POPA — Pop all registers off stack

PUSH immediate — Push immediate number on stack

IMUL destination register, source, immediate — Immediate x source to destination

SHIFT/ROTATE — Shift register or memory contents specified immediate destination, immediate number of times

Intel 80286

Salient features of 80286

- High performance microprocessor with memory management and protection
  - 80286 is the first member of the family of advanced microprocessors with built-in/on-chip memory management and protection abilities primarily designed for multi-user/multitasking systems
- Available in 12.5MHz, 10MHz & 8MHz clock frequencies

Salient features of 80286 :bus and memory sizes cont…

- The 80286 CPU, with its 24-bit address bus is able to address 16MB of physical memory.

- 1GB of virtual memory for each task

Microprocessor	Data bus width	Address bus width	Memory size
8086	16	20	1M
80186	16	20	1M
80286	**16**	**24**	**16M**

Salient features of 80286 Operating Modes

Intel 80286 has 2 operating modes:

- Real Address Mode :
  - 80286 is just a fast 8086 --- up to 6 times faster
  - All memory management and protection mechanisms are disabled
  - 286 is object code compatible with 8086
- Protected Virtual Address Mode
  - 80286 works with all of its memory management and protection capabilities with the advanced instruction set.
  - it is source code compatible with 8086

Salient features of 80286 cont…

- 286 includes special instructions to support operating system.
  - for example, one instruction can
    - i) ends the current task
    - ii) save its states
    - iii) switch to a new task
    - iv) load its states and
    - v) begin executing the new task
- housed in 68-pin package